Helium Step Heats on Geothite at Water SaturatedConditions Show Diffusive Helium Loss and High Retentivity at Surface Temperatures

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Abstract

(U–Th)/He dating of goethite (FeOOH) can directly date weathering, ferruginization, and laterite formation, but interpreting goethite He dates requires knowing the conditions under which goethite retains radiogenic helium. Vacuum step heats, the standard tool for characterizing helium retention, convert goethite to hematite and record dehydroxylation rather than diffusion (Farley et al., 2024). Because high water pressure suppresses dehydroxylation, we use the method described in (Thyer et al., 2026) to conduct the first helium step heat experiment on proton-irradiated goethite under water-saturated conditions. Goethite sample Win-06-01b (Flinders Ranges, South Australia) was encapsulated in a silica glass ampoule with distilled water and heated through multiple prograde and retrograde cycles between 170 and 250 °C over about 40 days. Apparent ^{3}He diffusivities are approximately seven natural log units lower than those measured on the same sample in vacuum and, beyond 7% cumulative release, define a linear Arrhenius array with an activation energy of 157.1 ± 5.3 kJ/mol and a closure temperature of 98.1 ± 8.6 °C at 10 °C/Ma. Reanalysis of two published vacuum step heats

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of the same sample shows that vacuum release follows first-order conversion to hematite, whereas the water-saturated release shows the history dependence of diffusion and follows multiple diffusion domain systematics. Thermodynamic and nucleation modeling calibrated on the vacuum data shows that the roughly tenfold smaller driving force for dehydroxylation at water saturation raises the barrier to hematite nucleation enough to suppress solid-state conversion on experimental timescales. Fluid-mediated conversion, observed in synthetic goethite pigment, and goethite recrystallization cannot be excluded; if either contributed, the diffusivities are upper bounds. A four-domain model places 97% of the ^{3}He in a single domain that is nearly quantitatively retentive at 25 °C over billion-year timescales. The ^{4}He/^{3}He spectrum nonetheless requires natural helium loss from this domain, which thermally activated diffusion can explain: steady cooling from 76–80 °C at 150–175 Ma fits best, while burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track data, also fits the ^{4}He/^{3}He spectrum. Water-saturated step heats enable systematic experiments on how crystallite size, composition, and radiation damage control helium retention in goethite.

Keywords: goethite, (U–Th)/He geochronology, helium diffusion, step heating, dehydroxylation, multiple diffusion domains

1. Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster et al., 2005; Farley et al., 2024). Goethite is a powerful chronometer for dating surface processes such as weathering and ferruginization di-

rectly: laterite and duricrust formation and recrystallization (Monteiro et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. As with any radiochronometer, meaningful geological interpretation of a measured age depends on knowing when the system is open to loss or gain of parent or daughter nuclides. For goethite, step heats of proton-irradiated, crystalline samples are used to reveal the degree of partial helium loss by the 4He/3He method (Shuster et al., 2005), and suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved, because goethite decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1999; Farley, 2000, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster et al., 2006; Guenthner et al., 2013). This approach

has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was partially lost in nature. The results of Shuster et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and

the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to measure helium loss (Shuster et al., 2005; Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

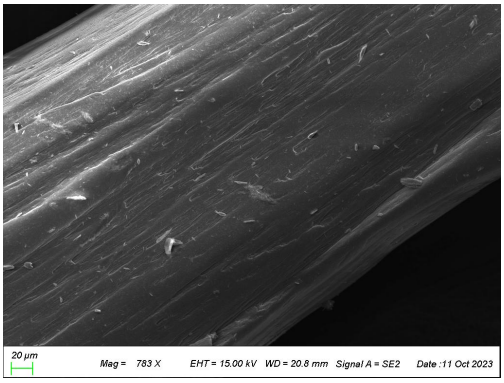
Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly contradicted by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (Vasconcelos et al., 2013). However, the models presented by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the

(U–Th)/He method and require experimental verification.

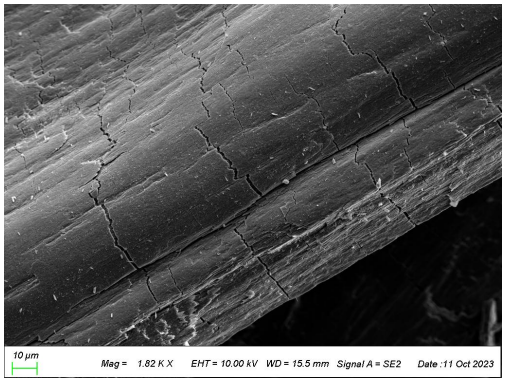
Goethite dehydration is suppressed at high water pressure (Langmuir, 1971; Navrotsky et al., 2008), so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature.

Early trials of the method were encouraging (§2.4). SEM imagery taken of thermally treated natural goethites in vacuum and open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Fig. 1b,d), while goethite heat-treated when encapsulated with water did not form cracks (Fig. 1c). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

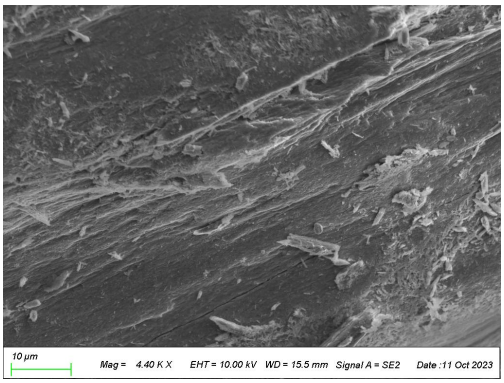
We test this premise with the first helium step heat experiment on goethite under water-saturated conditions, and compare it with published vacuum step heats of the same sample (§4.1–4.2). A thermodynamic and kinetic analysis of goethite dehydroxylation then explains why water has this effect (§2.3–4.4).



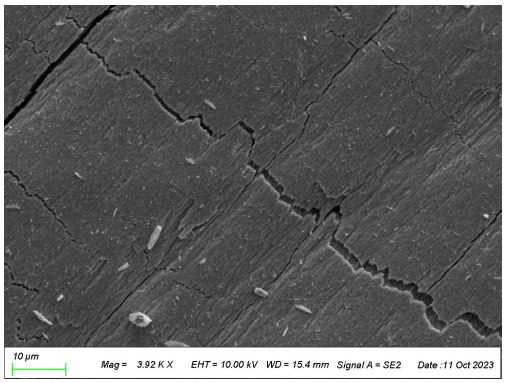
(a) Control



(b) Vacuum, 300°C, 11 h



(c) Water-buffered, 300°C, 4 h



(d) Vacuum, 300°C, 11 h

Figure 1: Secondary electron images of Bahia goethite aggregate before and after heat treatment at 300°C. (a) Untreated control grain; it has some rough surfaces and rare long-axis-parallel cracks. (b) After 11 hours in vacuum, cracks developed both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite. (c) After 4 hours in a sealed glass tube with water, no cracks indicative of dehydration were visible; most of the aggregate's surface was smoother than the area pictured, which had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some dehydration had occurred, or it may have been inherited from before hydrothermal treatment, as in the control. (d) The vacuum-heated grain of (b) at the same scale as (c), showing rough cracks at various scales both parallel and orthogonal to the long axis; these are the typical decomposition features that would be expected in (c) if significant decomposition had occurred there.

2. Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in South Australia (Waltenberg, 2012; Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005) and subsequently was subjected to helium step heat measurements in vacuum by Monteiro (2017) and Farley et al. (2024). In their step heat experiment of Win-06-01b, Farley et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley et al. (2024) clearly agree well with Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by Monteiro (2017) and Farley et al. (2024) was encapsulated with MilliQ water within a silica glass tube following Thyer et al. (2026). Fig. 2 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about 3×10^4 torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.



Figure 2: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of the thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would have been hotter than the recorded temperature, which would not affect the conclusions of this work.

Helium isotope measurements of evolved gases from hydrothermal step heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak

hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He

and 4He contents immediately before the measurement.

Helium released by the sample reaches the mass spectrometer only after permeating the ampoule wall, so each measured step yield lags the true release. Step yields were corrected for this permeation lag with the bounding method of Thyer et al. (2026), which gives lower-bound, central and upper-bound values for every step (Thyer, 2026). The correction and the ampoule permeabilities measured during the experiment are described in the Supplementary Information (§S1), together with the diffusivity inversions, the isotope-ratio uncertainties and the kinetic analysis of the published vacuum step heats (§S2).

2.1. Kinetic and Diffusive Reductions of Step-Heat Data

We reanalyzed the published vacuum step heats of irradiated Win-06-01b by Farley et al. (2024) (hereafter *KAF*) and Monteiro (2017) (hereafter *HDM*), together with the water-saturated step heat of this study, by reducing every step both as a first-order conversion rate constant and as an apparent diffusivity. For the kinetic reduction, ^{3}He is assumed to be immediately released from the goethite lattice upon conversion to hematite, except for a fraction ϕ that is inherited by the hematite product. The fraction of helium released in each step is then $F_i = (1-\phi)\,\alpha_i$, where α_i is the fraction of goethite converted in step i.

We describe the kinetics of goethite dehydroxylation with a first-order rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A\exp\!\left(-\frac{E_a}{RT}\right), \tag{1}$$

For a schedule of isothermal steps of duration Δt_j at T_j, Eq. (1) integrates

to

$$\alpha_i = 1 - \exp\left[-A \sum_{j \le i}, \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right)\right], \tag{2}$$

Inverting Eq. (2) step by step gives an apparent rate constant for every step that is independent of any fitted parameter except ϕ:

$$k_i = -\frac{1}{\Delta t_i} \ln \frac{1 - \alpha_i}{1 - \alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1 - \phi}. \tag{3}$$

For comparison, the diffusion parameter of each step is obtained with the same Fechtig and Kalbitzer (1966) inversion for a sphere that is applied to the hydrothermal experiment (§S2.1).

Additionally, in both experiments release stalls at $F \approx 0.9$. The fitted $\phi = 0.101$ is the same for KAF and HDM. In HDM, the remaining ~10% is released slowly between 340 and 500 °C and on fusion. We interpret this fraction as ^{3}He inherited by the hematite product and subsequently released by the much slower diffusion out of hematite. The conversion and the plateau correspond to the "Stage 2" and "Stage 3" release of Farley et al. (2024), whose transition near $F \approx 0.9$ is independent of heating schedule (and tracked by a distinct shift in ^{4}He/^{3}He: Fig. 9). The kinetic inversions (Eq. (3)) for these latest steps are highly sensitive to the choice of ϕ and are excluded from the fits of the kinetic parameters. The KAF steps run with the trap isolated are also excluded from the Arrhenius regression.

2.2. Multiple Diffusion Domain and Thermal History Modeling

Together, the ^{3}He release and the bulk-normalized ^{4}He/^{3}He spectrum constrain the thermal history, if the activation energy for diffusion is known (Farley and Flowers, 2012). These data from HT-Win-06-01b are interpreted

in two stages. First, the diffusion domain structure of the sample is inverted from the ^{3}He release alone. Second, natural thermal histories are fitted to the ^{4}He/^{3}He spectrum with that domain structure held fixed, so that every thermal history must work within the bounds set by the ^{3}He data. Derivations and numerical details are given in the SI (§S6).

Diffusion domain structure. The sample is represented by N non-interacting spherical domains that share the activation energy and frequency factor of the Stage 1 regression (§3.1). Domain j holds a fraction f_j of the uniformly distributed, proton-induced ^{3}He ($\sum_j f_j = 1$) and has radius r_j relative to the reference radius r_0 implied by the Stage 1 regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\!\left(-\frac{E_a}{RT}\right). \tag{4}$$

A model with N domains has $2N - 1$ free parameters. During the heating schedule, the reference domain accumulates the non-dimensional time $\tau_{0,i} = \sum_{m \le i} (D_0/a_0^2) \exp(-E_a/RT_m)\, \Delta t_m$ by the end of step i, and domain j reaches $\tau_{0,i}(r_j/r_0)^{-2}$. With $F(\tau)$ the cumulative fraction released by an initially uniform sphere (Crank, 1979), the model cumulative and fractional ^{3}He yields are

$$\Sigma F_i = \sum_{j=1}^{N} f_j\, F\!\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{5}$$

Following Farley and Flowers (2012), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_i \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_i \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{6}$$

summed over the 28 steps with more than 0.1 pccSTP of ^{3}He. The gas released in the six low-signal steps still counts toward the cumulative yields

of later steps, and their durations toward the heating schedule. Domains are added one at a time, each search starting from the $(N-1)$-domain solution, until an added domain no longer changes the fit or the domain structure. The Arrhenius and delta plots are not fitted: they follow from applying the Fechtig and Kalbitzer (1966) inversion to the model cumulative release, and their misfit is reported as a reduced χ^2 over the same 28 steps.

Thermal histories. Any thermal history predicts a ^{4}He/^{3}He spectrum that is obtained following the procedure of Shuster et al. (2004). A summary of this procedure is given in the SI (§S6.1). Thus, we evaluate how well each thermal history reproduces the measured ^{4}He/^{3}He spectrum. We compare three types of thermal histories, with two free parameters for steady cooling and the isothermal hold and three for burial and exhumation.

- **Steady cooling:** the temperature falls linearly from T_0 at t_{form} to 25 °C today. The goethite forms at an elevated temperature, for example from warm fluids at depth, and cools monotonically. Free parameters: t_{form} and T_0.

- **Burial and exhumation:** the path passes through (t_{form}, T_s), $(t_{\text{peak}}, T_{\text{max}})$, (50 Ma, T_s) and $(0, T_s)$ with $T_s = 20\,°\text{C}$. For this model, we assume that the goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, roughly following the thermal history (Gibson and Stüwe, 2000; Mitchell et al., 1998) interpreted for the region from apatite fission track thermochronometry. We fit t_{peak} and T_{max} at each of three fixed crystallization ages of 180, 195 and 210 Ma; because the crystallization age is also varied across the

three fits, burial has three free parameters.

- **Isothermal hold:** $T = T_{\mathrm{iso}}$ from t_{form} to the present. Free parameters: t_{form} and T_{iso}.

Each path is scored against

$$\chi^2 = \sum_i \left(\frac{R_i^{\mathrm{mod}} - R_i^{\mathrm{M}}}{\sigma_i} \right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2} \left(R_i^{\mathrm{UB}} - R_i^{\mathrm{LB}} \right), 0.02 \right], \tag{7}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of ^{3}He. Where M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method (Thyer, 2026). This analysis fits 24 independent data points; subtracting the fit parameters gives $\nu = 22$ degrees of freedom for steady cooling and the isothermal hold and $\nu = 21$ for burial and exhumation, which are used to evaluate the reduced χ^2. χ^2 is evaluated for these free parameters along regularly spaced grids to produce a likelihood heatmap (Fig. 10). The measured He age (130–180 Ma; Waltenberg 2012) is treated as a hard constraint. The best fit of each family is the lowest-χ^2 path with a predicted age in that range, and parameter ranges are those of the age-consistent paths within the same $\Delta\chi^2$ of that best fit.

2.3. Thermodynamic and Nucleation Model of Goethite Dehydroxylation

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\mathrm{fluid})}, \tag{8}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\mathrm{hem}}(T,P) + G_{\mathrm{w}}(T,P) - 2\,G_{\mathrm{goe}}(T,P) + \Delta G_{\mathrm{surf}}. \tag{9}$$

where ΔG_r is the Gibbs energy change of the reaction and G_{hem}, G_{w}, and G_{goe} are the Gibbs energies of hematite, water, and goethite, respectively. ΔG_{surf} is the surface energy contribution, which must be considered to reproduce observed phase stability (Navrotsky et al., 2008). Goethite is stable where $\Delta G_r > 0$ and hematite plus water where $\Delta G_r < 0$. The values and equations used to evaluate each of these terms are given in the supplement: §S4.

Nucleation barrier. In classical nucleation theory (Gibbs, 1878; Volmer and Weber, 1926; Becker and Döring, 1935), reviewed by Christian (2002), Kashchiev (2000) and, for mineral systems, De Yoreo and Vekilov (2003) and Lasaga (1998), a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{\mathrm{hem}}} + 4\pi r^2 \gamma, \tag{10}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table 1), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Nuclei larger than the critical radius $r^* = 2\gamma V_{\mathrm{hem}}/\Delta\mu$ grow, and the energetic barrier to forming one is

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{\mathrm{hem}}^2}{3\,\Delta\mu^2}. \tag{11}$$

In the first-order limit of §4.1, the rate constant is a nucleation rate of the form derived by Turnbull and Fisher (1949),

$$k = A\exp\left(-\frac{E_a}{RT}\right)\exp\left(-\frac{\Delta G^*}{k_B T}\right),$$
$$\frac{\Delta G^*}{k_B T} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{\mathrm{hem}}^2}{3k_B} = 1.112\times 10^{15}\ \mathrm{m^6\,K\,J^{-1}\,mol^{-2}}, \tag{12}$$

where $A\exp(-E_a/RT)$ describes the atomic mobility and $\exp(-\Delta G^*/k_B T)$ is the equilibrium fraction of sites occupied by critical nuclei. Because the

barrier scales as $\gamma^3/\Delta\mu^2$, the tenfold drop in $\Delta\mu$ between vacuum and water saturation raises the energetic barrier ΔG^* a hundredfold (Table 1).

Internal nucleation. Goethite dehydroxylates topotactically: hematite nucleates within the parent crystal, inherits its oxygen framework and orientation, and the pseudomorph develops pores and twinned hematite domains (Lima-de Faria, 1963; Watari et al., 1979, 1983; Cudennec and Lecerf, 2005). We therefore consider only internal nucleation, in which the new interface of the burgeoning nuclei is between hematite and goethite. We know of no measured or calculated energy for that interface, so γ is replaced by an effective parameter γ_{eff}, calibrated from the data.

2.4. SEM Study of Thermally Treated Goethite

In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster et al., 2005; Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley et al. (1999) and in sealed glass ampoules with water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech in secondary electron mode by the ZEISS 1550VP Field Emission SEM.

3. Results

3.1. Water-Saturated Step Heat of HT-Win-06-01b

The diffusivities determined from the step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and follow evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follows a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreases in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 3, Fig. 4). The Arrhenius plot of the 4He diffusivities differs in shape: it is concave-down to about 5% cumulative 3He yield, after which it follows a roughly linear array about one log unit below the 3He array (Fig. 3). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 5b). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are within error of Delta = 0: the reference line (Fig. 4). A series of low-yield measurements

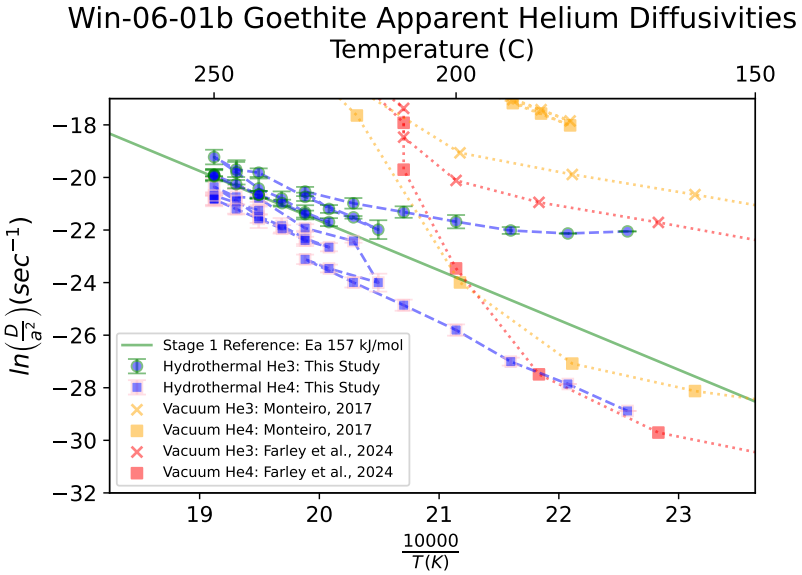


Figure 3: Linear regression of the last 12 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He; however, they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

on the first retrograde cycle produce high-error, scattered diffusivities which are omitted for clarity.

3.2. Kinetic Reanalysis of Vacuum Step Heats

All steps are plotted in Fig. 5, for both the kinetic and diffusion reductions, but the fits use only KAF steps 5–22 (185–280 °C) and HDM steps 8–26 (180–296 °C).

An unweighted least-squares fit of $\ln k_i$ against $1/T_i$ for the 35 steps in the fitting window gives

$$E_a = 137.0 \pm 2.7 \text{ kJ mol}^{-1}, \qquad \ln A = 21.91 \pm 0.66 \tag{13}$$

KAF alone gives 139.3 ± 6.2 and HDM alone 135.4 ± 2.5 kJ mol^{-1}), in agreement with the cumulative fit. The result does not depend on the fitting window: fitting all steps instead gives $E_a = 122.7$ kJ mol^{-1} and the same ranking of models (§S3).

The kinetic rate parameters fitted to KAF alone predict the independent HDM schedule with an rms misfit in F of 0.018 over the fitting window. A single-domain spherical diffusion model fits both vacuum step heats worse, even when given the same hematite-inherited fraction ($\phi = 0.101$) as the first-order fit (Table S2).

KAF steps 1–4 and HDM steps 2–7 (below ∼180 °C) release ^{3}He 0.5–3 orders of magnitude faster than the extrapolated conversion rate (Fig. 5). This component amounts to less than 2% of the ^{3}He.

In two KAF steps the trap was isolated and evolved water accumulated in the line. At 230 °C the step rate constant fell to 0.33 times the mean of its neighbors at the same temperature, and at 240 °C to 0.22 times. The rate

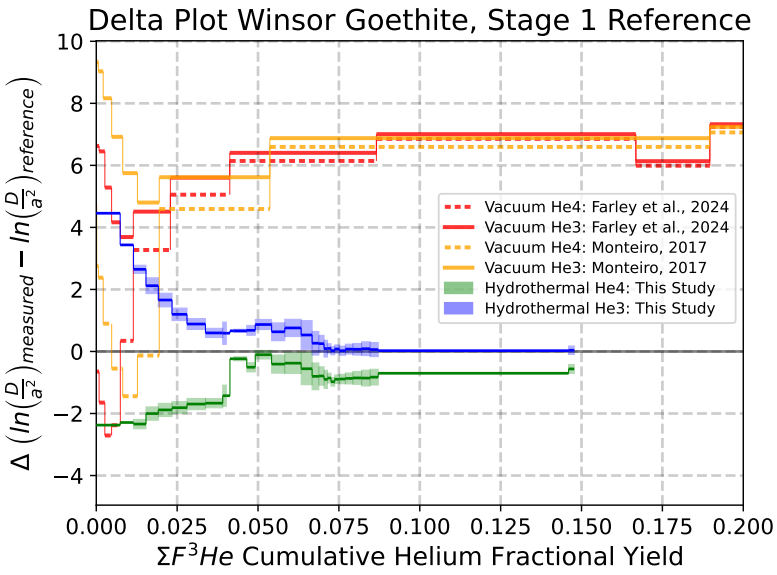


Figure 4: Delta plot similar to those presented in Farley (2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. Six measurements from the water-saturated experiment are omitted for clarity due to low signals and high errors. For visual continuity, the gap left behind is bridged and no uncertainty is shown. 3He monotonically decreases until settling on the reference line from 7–15% 3He fractional yield. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in variably retentive domains. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

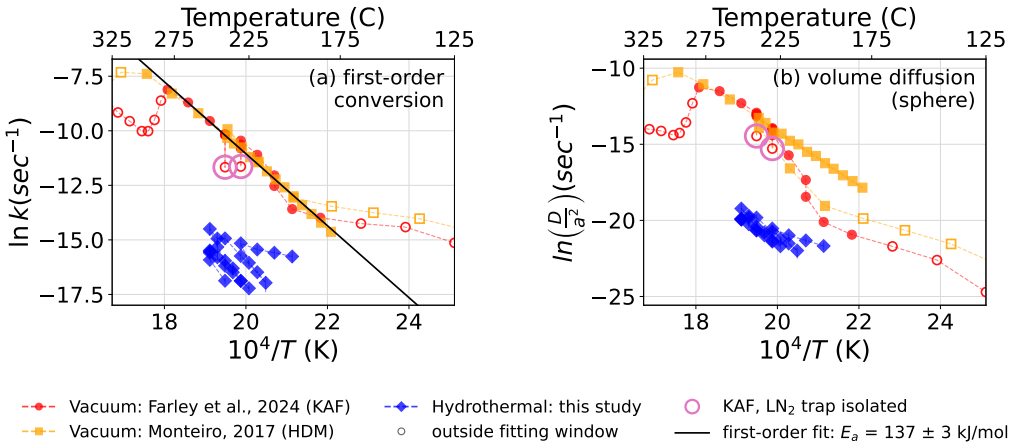


Figure 5: The ^{3}He release of every step of the two vacuum step heats of Win-06-01b (red circles: KAF, Farley et al. 2024; orange squares: HDM, Monteiro 2017) and of the hydrothermal step heat of this study (blue diamonds), reduced in two ways: (a) as first-order conversion rate constants k_i (Eq. (3), $\phi = 0.101$), and (b) as apparent diffusivities D/a^2 for a sphere (Fechtig and Kalbitzer, 1966). Only the filled points are used for the linear regressions. Large pink circles mark the two KAF steps with the LN_2 trap isolated. Six low-signal steps of the water-saturated experiment are excluded for clarity. The black line is a first-order Arrhenius fit to filled points of both vacuum experiments. These fitted parameters are given in Eq. (13). Below ~180 °C both vacuum experiments depart from the first-order line. Steps with $F > 1 - \phi$ have no defined k and are absent from (a).

recovered fully when the trap was restored. Water and ^{3}He were suppressed together. We return to these two steps in §4.4, where they calibrate the nucleation model.

4. Discussion

4.1. Dehydroxylation Kinetics in Vacuum Step Heats of Win-06-01b

Farley et al. (2024) showed that evolved helium in vacuum step heats of Win-06-01b (and many other goethite samples) is controlled by the conversion of goethite to hematite. Here, we present a reanalysis of the vacuum step heat data of Win-06-01b from KAF and HDM to show that the helium release of Win-06-01b is better described by a model of first-order reaction kinetics for goethite dehydroxylation than by a second order diffusion model.

We postulate that the early ^{3}He released faster than the extrapolated conversion rate (§3.2) is diffusively lost from the least retentive domains; it corresponds to "Stage 1" helium release of Farley et al. (2024). The kinetic reduction yields simpler, more interpretable Arrhenius plots for the vacuum experiments than the diffusion inversion does. The kinetic inversion collapses the prograde steps at the beginning of the vacuum experiments onto a single line that the diffusion reduction instead shows as an irreversible increase in diffusivity (Fig. 5). This initial helium release is most easily read as a diffusion-related loss from a broad distribution of small, poorly retentive domains in the goethite. The Arrhenius plot of the kinetic reduction makes it clear that helium release from the initial steps is in excess of the rate of dehydroxylation, which is not obvious from the Arrhenius plot produced with the diffusion reduction. This shift in the dominance from diffusive loss

to dehydroxylation-related loss offers a clear explanation for the change in behavior between "Stage 1" and "Stage 2" in vacuum step heat experiments described by Farley et al. (2024).

Thermogravimetric studies of goethite dehydroxylation report somewhat higher apparent activation energies, 154±15 to 169±8 kJ mol^{-1}, for a reaction controlled by a two-dimensional phase boundary, with a different mechanism at early stages and low temperature (Goss, 1987). This may simply be due to a difference in samples: goethites from different sources have different particle sizes, morphologies, and aluminum contents, all of which affect the kinetics of dehydroxylation (Langmuir, 1971; Yapp, 1983; Tardy and Nahon, 1985; Majzlan and Navrotsky, 2003).

4.2. Helium Release under Water-Saturated Conditions Is Diffusive

The goal of this work was to attempt to slow the goethite dehydroxylation process below the rate of thermal helium loss in unchanging goethite. The thermal schedule included several prograde-retrograde heating cycles which together would have completely dehydroxylated goethite in vacuum, using the vacuum kinetics of Eq. (13). Even so, only 14.7% of the ^{3}He was released in the water-saturated experiment; we argue below that the helium loss in the water-saturated experiment is controlled by diffusion rather than by conversion.

Fig. 5 compares the kinetic and diffusion inversions of the step heat data. These plots show the opposite trend for the vacuum and hydrothermal experiments: the hydrothermal experiment collapses into a simpler, coherent linear array with the diffusivity reduction, whereas the kinetic inversion of the same data records a continuously declining k for any given temperature

as the experiment proceeds, forming a zig-zag pattern. This is the expected behavior for a second-order (diffusion-controlled) process for which helium loss slows more strongly than a first-order process as the experiment proceeds. Fig. 5 shows that helium loss is first-order for vacuum experiments and second order for the water-saturated experiment. Section 4.4 shows that these observations require the conversion rate at water saturation to be at least $10^{2.5}$ times slower than in vacuum, and that a nucleation control on the dehydroxylation kinetics explains suppression far greater than that.

While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the results presented here suggest that a second-order process (likely volume diffusion) is the dominant mechanism of helium loss for Win-06-01b goethite in water-saturated conditions.

4.2.1. Consistency with MDD Systematics

All existing helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (Monteiro, 2017; Farley et al., 2024), which likely could follow MDD systematics (Farley, 2018). In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, or for a system where the activation energy of diffusion is known, the temperature dependence can be removed by referring to the data in relation to a temperature-dependent reference line, as in delta or r/r0 plots (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018). For the activation energy of 157.1 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step

heat data follow the expected MDD systematics and monotonically decrease within a delta plot (Fig. 4), which is consistent with volume diffusion in an MDD system.

The strong plateau in the delta plot of the water-saturated goethite experiment and the statistically robust regression over a wide range of fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggest that the frequency factor obtained by the regression may be dominated by a single diffusion domain for $F_c \geq 7\%$ rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018).

A multiple diffusion domain inversion of the ^{3}He release (§2.2) tests this claim (Table S5; Fig. 6). A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 27$), and two domains capture the high early diffusivities but leave a poor Arrhenius fit ($\chi^2_\nu = 3.3$). Three domains reduce the yield misfit by more than half ($\chi^2_\nu = 1.08$). With four, the smallest domain splits in two and the fit improves slightly further ($\chi^2_\nu = 1.00$). Every domain added beyond four coincides in size with the dominant domain and leaves the fit unchanged. We therefore adopt four domains as a complete description of the ^{3}He data:

- 0.57% of the ^{3}He at $r/r_0 = 6.8 \times 10^{-4}$;
- 0.66% at $r/r_0 = 2.9 \times 10^{-3}$;
- 1.5% at $r/r_0 = 0.031$;
- 97.2% at $r/r_0 = 1.12$.

The three small domains, holding 2.8% of the ^{3}He in total, are fully degassed during the experiment and account for the high early diffusivities and the

gradual approach of the delta plot to the Stage 1 line (Fig. 7). The dominant domain releases 12% of its ^{3}He during the experiment and diffuses at 0.80 times the Stage 1 rate. The Stage 1 line is therefore a slightly fast description of the dominant domain, because the bulk-inverted diffusivities on which it was regressed still carry a small contribution from the fast domains. The four-domain model reproduces the Arrhenius array, including its retrograde segments, and the delta plot, although neither is fitted directly. Fits for $N = 1$–5 are shown in the SI (Fig. S5).

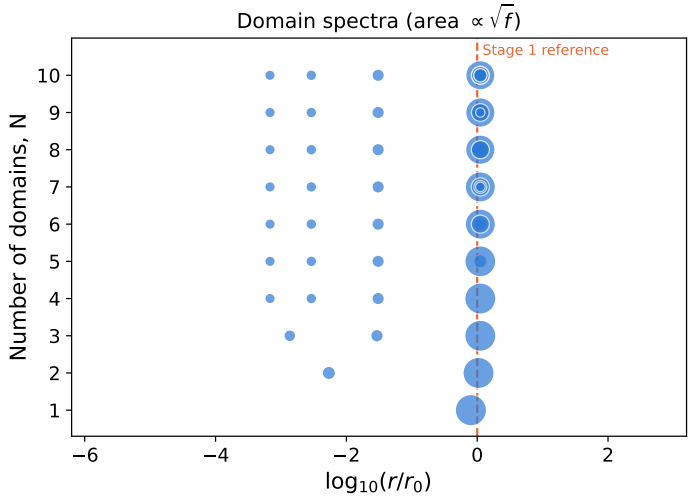


Figure 6: Domain spectra of the best-fitting multiple diffusion domain models for $N = 1$–10. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

4.3. Goethite Stability and the Driving Force for Dehydroxylation

The preceding sections establish from helium step heat data alone that the vacuum step heats of Win-06-01b record first order conversion to hematite,

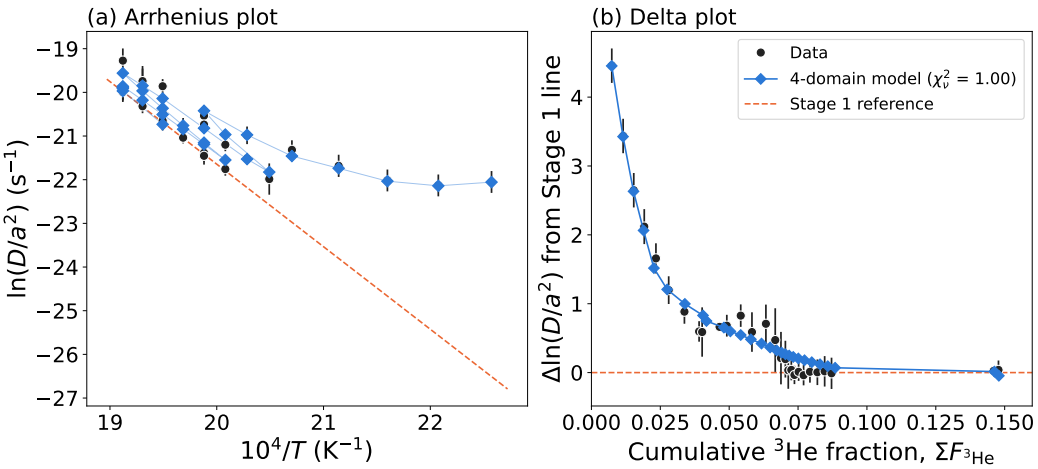


Figure 7: (a) Arrhenius plot and (b) delta plot (departure from the Stage 1 line) of the measured ^{3}He diffusivities of HT-Win-06-01b, with the values implied by the four-domain fit to the ^{3}He yields. Neither plot is fitted directly. The six low-signal steps are omitted.

whereas the water-saturated step heat records second order diffusion without detectable dehydroxylation. Here we use the thermodynamics of goethite dehydroxylation to explain why water has this effect and to estimate how much solid-state conversion could have occurred in the hydrothermal experiment (§4.4).

This model can be used to calculate the stability field for goethite with respect to dehydroxylation to hematite and water in log P_{H_2O}–T space (Fig. 8). This phase diagram communicates several important features of the goethite–hematite–water system that are relevant to the experiments considered here:

1. Goethite is not stable at the conditions of any of the experiments considered here, in water or in vacuum, in spite of the evidence that it did not react in the water-saturated experiment. Thus, we hypothesize that the reaction is kinetically inhibited for goethite under these

conditions.

2. The stability of goethite is strongly dependent on the partial pressure of water below and up to water saturation. Above the saturation pressure, water is liquid and contributes far less to the reaction volume than it does as a gas, becoming a secondary control on goethite's stability.
3. The stability of goethite is strongly dependent on grain size, with smaller grains being more stable than larger grains. At water saturation, bulk goethite is stable below 18 °C, 100 nm goethite below 28 °C, and 10 nm goethite below 108 °C.

Stability alone does not determine what happens in a step heat; the rate of conversion controls whether this equilibrium state is reached or if the reaction proceeds at all, and that depends on the chemical potential of the reaction: $\Delta\mu$. Table 1 gives the bulk driving forces at the experimental temperatures, evaluated with IAPWS-95 water. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line ($P_{H_2O} \approx 10^{-8}$ atm) it is 76–106 kJ, about ten times larger. The chemical potential of water is therefore the key variable separating the two experimental settings.

4.4. Water Pressure Control on the Hematite-in-Goethite Nucleation Barrier

The hydrothermal data bound how fast goethite could have converted to hematite in water. Under the vacuum kinetics of Eq. (13), the hydrothermal schedule gives $\sum_i k_{vac}(T_i)\,\Delta t_i \approx 60$, enough to convert the sample completely. If water lowers the conversion rate by a factor $R = k_{sat}/k_{vac}$, then even if all of the observed ^{3}He release came from conversion ($\alpha = F/(1-\phi) = 0.147/0.899 = 0.16$), R can be at most $10^{-2.5}$. The release is second order

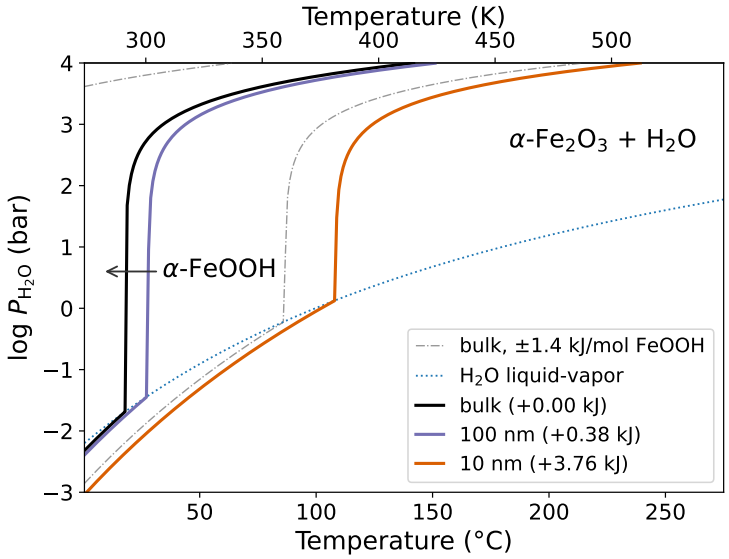


Figure 8: Goethite–hematite–water equilibrium in $\log P_{H_2O}$–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (S8)). Goethite is stable above and to the left of each curve. The dash-dot curves illustrate the absolute uncertainty: it is the bulk stability boundary shifted by the ± 1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan et al. (2003). The blue dotted curve shows the water vapor saturation pressure; liquid water is present at and above this curve. The thermodynamic data used to compute the phase diagram are in Table S3.

Table 1: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation and at $P_{H_2O} = 10^{-8}$ atm.

T (°C)	P_{sat} (bar)	$\Delta\mu_{sat}$	$\Delta\mu_{sat}/RT$	$\Delta\mu_{vac}$	$(\Delta\mu_{vac}/\Delta\mu_{sat})^2$
150	4.76	5.68	1.61	75.8	178
175	8.93	6.88	1.84	83.4	147
200	15.55	8.11	2.06	91.0	126
225	25.50	9.37	2.26	98.5	111
250	39.76	10.66	2.45	106.0	99

volume diffusion (§4.2). Limiting the dehydroxylation fraction to below 1% requires $R \leq 10^{-3.8}$. Here we show that classical nucleation theory, calibrated on the vacuum data alone, predicts an even stronger rate suppression under water-saturated conditions.

Calibration. When Farley et al. (2024) isolated the liquid nitrogen trap from the heated goethite, they raised P_{H_2O} in the line and lowered $\Delta\mu$, and the isothermal rate fell by a factor s (§4.1). With A and E_a unchanged, Eq. (12) gives

$$\gamma_{\text{eff}} = \left[\frac{T \ln(1/s)}{C\left(\Delta\mu_{\text{off}}^{-2} - \Delta\mu_{\text{on}}^{-2}\right)} \right]^{1/3}, \tag{14}$$

with $\Delta\mu_{\text{on}}$ evaluated at $P_{H_2O} = 10^{-8}$ atm and $\Delta\mu_{\text{off}}$ at an upper bound on the water pressure with the trap isolated. This pressure estimate of the water released depends on the volume of the experimental apparatus and the water content emitted by the goethite, which was measured with capacitance manometry (§S5). This gives $\gamma_{\text{eff}} = 0.16$ J m^{-2} (range 0.14–0.19). Under vacuum the corresponding critical nucleus is smaller than a formula unit ($r^* = 0.08 - -0.11$ nm) and the barrier is only 0.5–1.3 k_BT: conversion in vacuum is effectively barrierless, consistent with the simple Arrhenius behavior of both vacuum experiments. At this scale classical nucleation theory has no literal meaning, so γ_{eff} is best read as the single parameter of a rate law in which $\ln k$ falls as $-1/\Delta\mu^2$, not as an interface energy. At saturation, $r^* \approx 1$ nm and the barrier is 51–129 k_BT.

Translation to water saturation. For the same sample, only $\Delta\mu$ changes between vacuum and saturation:

$$\frac{k_{\text{sat}}}{k_{\text{vac}}} = \exp\left[-\frac{C\gamma_{\text{eff}}^3}{T} \left(\frac{1}{\Delta\mu_{\text{sat}}^2} - \frac{1}{\Delta\mu_{\text{vac}}^2} \right) \right]. \tag{15}$$

Applied step by step to the hydrothermal schedule, Eq. (15) predicts a total conversion ($\sum_i \frac{k_{\mathrm{sat}}}{k_{\mathrm{vac}}}(T_i)\, k_{\mathrm{vac}}(T_i)\, \Delta t_i$) of $10^{-20.5}$, $10^{-31.4}$ and $10^{-53.7}$ for $\gamma_{\mathrm{eff}} =$ 0.14, 0.16 and 0.19 J m^{-2}, and $10^{-67.3}$ to $10^{-15.7}$ (95%) when the uncertainties in the calibration and in the thermodynamic data are propagated (§S5). The exact values reflect the assumed $\Delta\mu^{-2}$ dependence; what matters is that the model predicts suppression far beyond the $10^{-3.8}$ that the data require.

Assumptions. This is a consistent physical explanation of the data, not a proof. It is not exhaustive. This analysis assumes (i) that the trap-isolated slowdown acts through $\Delta\mu$, rather than, for example, through water blocking reactive sites; (ii) that γ_{eff}, A and E_a are the same in vacuum and in water; (iii) that the barrier depends on the bulk driving force without significant contribution from surface terms (this analysis extended to 10 nm grains raises the predicted conversion only to $\sim 10^{-23}$; §S5); and (iv) that conversion is solid-state. Conversion by dissolution of goethite and precipitation of hematite from the fluid is not described by Eq. (15) (see §S7). The calibrated γ_{eff} absorbs contributions from the sample's microstructure and cannot be transferred to other goethites.

4.5. Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments

The aim of helium step heat experiments is to characterize the mechanisms and rates of helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. The results of this experiment are encouraging in that they differ remarkably from experiments on the same samples in vacuum that are primarily controlled by dehydration processes (Farley et al., 2024), but we cannot exclude that this experiment

may have been affected to some degree by processes aside from volume diffusion that are unrelated to helium loss in goethite in nature. We consider two processes outside of volume diffusion that could have contributed to the helium released in the water-saturated experiment: (i) recrystallization of goethite or (ii) fluid-mediated conversion of goethite to hematite. If either of these processes affected the helium release, then the apparent diffusivities reported here would represent upper bounds for diffusive helium loss in goethite in nature.

The helium step heat data at water-saturated conditions reported here are better fit with a diffusion model than with a conversion model (§4.2). The kinetic analysis of §4.1 and the nucleation analysis of §4.4 indicate that topotactic dehydroxylation has a nucleation barrier that is insurmountable for experimental conditions and timescales; however, this model has no bearing on fluid-mediated dehydroxylation. Equilibrium thermodynamics predict that goethite is unstable with respect to hematite and water at these conditions (Majzlan et al., 2003) (§4.3) and therefore dehydroxylation of goethite in the experiment cannot be outright excluded.

The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for many days. These conditions appear to have visibly dissolved and reprecip-

598 itated silica from the ampoule's inner walls (Fig. 2), qualitatively testifying
599 to the ability of the hydrothermal fluid to dissolve and recrystallize rock-
600 forming oxides. It is unquestionable that recrystallization of some goethite
601 could have occurred under these conditions; however, it is unclear if condi-
602 tions were favorable to recrystallize this specimen of goethite, Win-06-01b,
603 which is amongst the most helium-retentive and crystalline dated goethite
604 samples (Waltenberg, 2012; Farley et al., 2024). Less crystalline goethites in
605 nature are commonly recrystallized, which appears to result in helium loss,
606 resetting the U-Th/He age (Monteiro et al., 2018, 2014; Heim et al., 2006).
607 Recrystallization could be the dominant process controlling helium loss in
608 this experiment, but only if it is faster than diffusive loss, in which case the
609 data presented here would represent new upper bounds for apparent diffu-
610 sivities for helium in Win-06-01b goethite. Recrystallization might also be
611 faster than volume diffusion of helium in goethite in nature over geologic
612 timescales even if it did not occur in this experiment. In that case, recrys-
613 tallization could be the primary driver of helium loss in nature regardless of
614 the results of this experiment.

615 Synthetic goethite pigment heated with excess water converted to hematite
616 by dissolution and reprecipitation (SI, §S7), showing that water-saturated
617 conditions do not by themselves exclude fluid-mediated conversion of goethite,
618 even where topotactic conversion is suppressed (§4.4). Whether that path-
619 way operates likely depends on the solution chemistry. Still, the helium step
620 heat data for Win-06-01b-HT show no evidence for the abrupt and extreme
621 helium loss that would be expected to accompany a fluid-mediated reaction.
622 The step heat data are instead consistent with a smooth, history-dependent

release of helium, as expected for diffusive loss (§4.2).

4.6. High Helium Retentivity of Win-06-01b-HT

The apparent diffusivities of the water-saturated experiment are approximately seven natural log units lower than those obtained from the same sample in vacuum by Monteiro (2017) and Farley et al. (2024) (Fig. 5b, Fig. 3, Fig. 4). This high helium retentivity is explored through three lenses: closure temperature of cooling samples (Dodson, 1973), isothermal diffusion (Wolf et al., 1998), and through comparison with step heats of other polycrystalline samples.

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. The multiple diffusion domain inversion (§4.2.1) indicates that the dominant domain, which holds 97% of the ^{3}He, diffuses at 0.80 times the rate of the Stage 1 line. Its closure temperature is correspondingly about 1.6 °C higher, well within the quoted uncertainty; the Stage 1 parameters are used throughout this section.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Monteiro et al., 2014; Hofmann et al., 2017; Allard et al., 2018). For crystals of goethite grown and collected near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973).

In such cases, a better measure of retentivity is an analysis of isothermal retentivity over time (Wolf et al., 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT yields t'/t = 0.968, where t' and t are helium age and isothermal holding time, respectively. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales.

This study of a single sample has no bearing on potential complexities related to aluminum content, radiation damage, or crystal defects on bulk diffusion (Bassal et al., 2022), nor does it permit estimation of the diffusivity at infinite temperature (D0) which would require experiments on controlled grain sizes.

4.7. Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT

Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples measure the rounding of concentration profiles of radiogenic helium by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster et al., 2004; Farley et al., 2024). Aside from the first experimental step, bulk-normalized ^{4}He/^{3}He ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 9). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley et al., 2024). These least retentive domains

could represent fine goethite grains, or they could represent interstitial helium trapped between crystallite grain boundaries. Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing 3He from goethite samples during step heats, and could potentially represent the LRD (low retention domain) of Shuster et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different conditions is known to have highly variable crystallinity, crystallite ordering, surface area, and non-stoichiometric water contents (Schwertmann et al., 1985).

The hydrothermal experiment records lower bulk-normalized ^{4}He/^{3}He ratios than the vacuum experiments at any given $\Sigma F_{^3\mathrm{He}}$ for the duration of the experiment. Similarly, the approach to the ^{4}He/^{3}He plateau varies between the three experiments: a high bulk normalized ^{4}He/^{3}He plateau is most decisively represented in the Farley et al. (2024) experiment, whereas the ^{4}He/^{3}He ratios in the step heats of Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in ^{4}He/^{3}He ratios is slowest for the hydrothermal experiment (Fig. 9). This suggests that

the geometry of helium loss differs between the hydrothermal and vacuum experiments, and could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest ^{4}He/^{3}He spectra are generated by higher water pressures. We interpret this experimental control on the ^{4}He/^{3}He spectra to reflect the relative importance of goethite dehydration and helium diffusion through crystalline goethite in governing the geometry of helium loss from the sample.

If the geometry of helium loss due to dehydration and due to diffusion differ, then the ^{4}He/^{3}He spectra of goethite measured in vacuum may not straightforwardly encode the thermal history of the sample in the way that such data faithfully encode this history in phases such as hematite or apatite (Farley and McKeon, 2015; Shuster et al., 2004). More ^{4}He/^{3}He spectra of goethite under hydrothermal conditions are needed to investigate this hypothesis further.

These ratios also bear on the natural thermal history of Win-06-01b. Throughout the range of $\Sigma F_{^3\mathrm{He}}$ that defines the Stage 1 reference line (7–15%), step ^{4}He/^{3}He ratios remain at 0.67–0.82 of the bulk value (Thyer, 2026). Radiogenic ^{4}He is depleted relative to the uniformly distributed ^{3}He in this domain, so even this retentive domain lost helium in nature. A domain with the Stage 1 parameters is nearly quantitatively retentive at Earth-surface temperatures (§4.6), so this loss is difficult to attribute to thermally activated diffusion at surface conditions. Either Win-06-01b experienced temperatures well above surface conditions (perhaps it grew in a hydrothermal setting or experienced burial heating), or non-thermal processes such

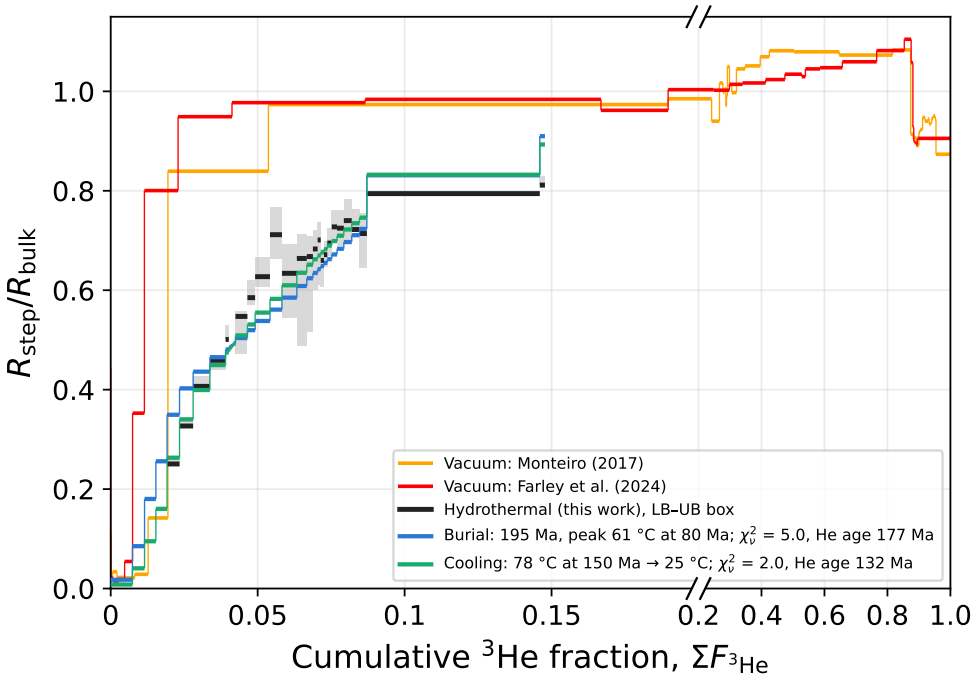


Figure 9: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b goethite step heats against cumulative fractional ^{3}He yield: the water-saturated hydrothermal experiment of this study (black, with bounding-method LB–UB boxes), the vacuum experiments of Monteiro (2017) (orange) and Farley et al. (2024) (red), and the best age-consistent thermal-history predictions fitted to the hydrothermal spectrum with the four-domain structure fixed (burial with crystallization at 195 Ma, blue; steady cooling, green; §4.8). The hydrothermal experiment shows lower bulk-normalized ^{4}He/^{3}He ratios at a given ΣF_{3He} for the duration of the experiment, suggesting that ^{4}He/^{3}He spectra are dependent on the partial pressure of water and that the geometry of goethite dehydroxylation differs from the geometry of helium diffusion through goethite. For all experiments, ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. Hydrothermal steps with less than 0.1 pccSTP of ^{3}He or without permeation-lag bounds are omitted. The horizontal axis changes scale at $\Sigma F_{3He} = 0.2$ (double slash) to emphasize the portion of the plot where hydrothermal data exist.

as fluid-mediated recrystallization caused partial helium loss in nature. Section 4.8 tests whether thermally activated diffusion alone, within the domain structure inferred from the ^{3}He release, can reproduce the measured spectrum.

4.8. Thermal History of Win-06-01b

We begin the exercise by limiting our exploration of potential thermal histories to geologically reasonable and simple paths. With the four-domain structure of §4.2.1 held fixed, we fitted steady-cooling, burial-and-exhumation, and isothermal thermal histories to the ^{4}He/^{3}He spectrum of HT-Win-06-01b (§2.2). Previous studies constrain our exploration of the burial-and-exhumation paths. Gibson and Stüwe (2000); Mitchell et al. (1998) interpret regional cooling from roughly 50–70 °C after the mid-Cretaceous from their apatite fission-track studies. Mitchell et al. (1998) additionally interpret, from limited 1.5 km thick sediment accumulation in Triassic intramontane basins, that parts of the ranges were near the surface in the early Mesozoic. Table S6 gives the best fit of each family, Fig. 9 compares the best steady-cooling and burial predictions with the spectrum, and Fig. 10 shows their χ^2 maps on a common color scale.

Steady cooling. Linear cooling to 25 °C fits the spectrum best. The best age-consistent path starts at 78 °C at 150 Ma ($\chi^2_\nu = 2.00$), and the age-consistent 1σ range is crystallization at 150–175 Ma at 76–80 °C, followed by cooling at $\sim$0.3 °C Myr^{-1}. Such paths pass through $\sim$55 °C at 90 Ma, inside the apatite fission-track window. The spectrum alone prefers somewhat younger, hotter paths: the unconstrained minimum ($\chi^2_\nu = 1.68$; 89 °C at 80 Ma) predicts

747 a He age of 68 Ma, and the unconstrained 1σ region lies entirely at ages
748 younger than 130 Ma (Fig. 10b).

749 *Burial and exhumation.* Burial paths fit the spectrum less well ($\chi^2_\nu = 5.0$–
750 6.5), but they agree with the independent regional constraints of Gibson
751 and Stüwe (2000) and Mitchell et al. (1998). For crystallization at 180 and
752 195 Ma, the best age-consistent paths peak at 61 °C at 70 and 80 Ma, with
753 1σ ranges of 58–65 °C between 98 and 60 Ma (Fig. 10a). These peaks fall
754 in or just after the apatite fission-track window, within the 50–70 °C range
755 of Gibson and Stüwe (2000) and at the <60 °C limit reported by Mitchell
756 et al. (1998). For crystallization at 210 Ma, only hotter and earlier peaks
757 (70–74 °C at 105–130 Ma) give an age within the measured range, and they
758 fit worse ($\chi^2_\nu = 6.5$).

759 *Isothermal hold.* The spectrum constrains only the product of diffusivity
760 and time, so the χ^2 minimum is a ridge from ∼47 °C at 50 Ma to ∼35 °C at
761 500 Ma, along which $\chi^2_\nu \approx 18.5$ throughout. No isothermal path reproduces
762 the shape of the spectrum: they underpredict the ratios of the early steps and
763 overpredict those of the late steps. We do not consider this family further.

764 *Where the families differ.* The steady-cooling and burial predictions differ
765 mainly in the earliest steps ($\Sigma F_{^3\mathrm{He}} \lesssim 0.035$; Fig. 9), which sample the outer
766 rim of the dominant domain. On the burial path the sample sits at 20 °C for
767 the last 50 Myr, long enough for radiogenic ^{4}He to re-accumulate in the rim,
768 so the early ratios are overpredicted. Steady cooling keeps the rim depleted
769 until geologically recent times, with diffusive loss declining monotonically
770 toward the present. Elsewhere both leave the same residual pattern: ratios

771 underpredicted between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.06, and overpredicted in the two
772 longest high-temperature steps.

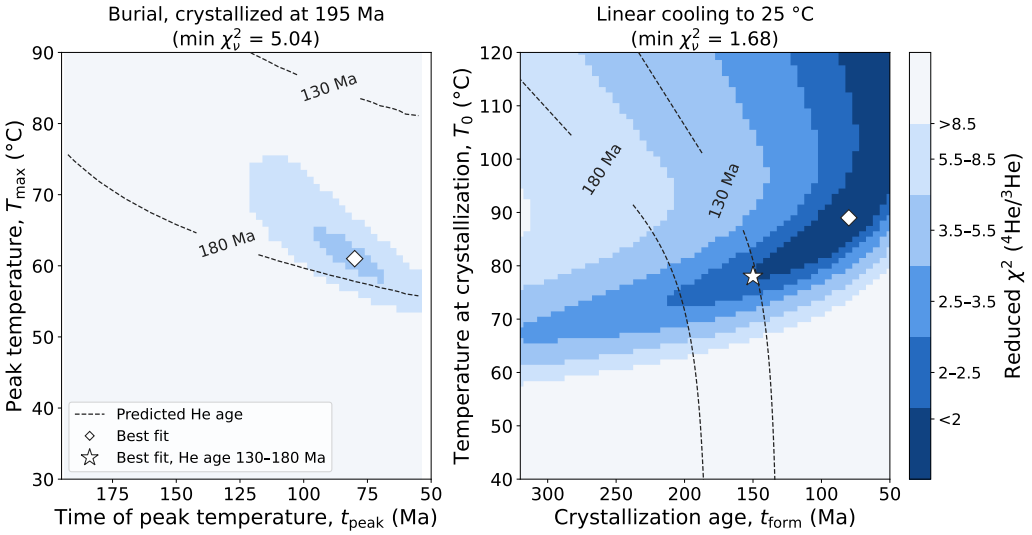


Figure 10: Reduced χ^2 of the ^{4}He/^{3}He spectrum for (a) burial paths with crystallization at 195 Ma, as a function of the time and magnitude of peak temperature, and (b) steady cooling to 25 °C, as a function of crystallization age and initial temperature. Both panels share one color scale, anchored to the overall minimum. Dashed lines show the predicted bulk He age at 130 and 180 Ma, diamonds the best fit, and the star the best fit with a He age of 130–180 Ma.

773 *Consistency with diffusive loss.* These two families of thermal paths encode
774 very different geologic origins. We note that any number of arbitrarily com-
775 plex thermal paths can be evaluated against the data with this workflow,
776 but a thorough search for viable thermal paths is beyond the scope of this
777 study, and is an ill-posed problem for a single sample. The main purpose of
778 this exercise is to demonstrate that the data we present here are consistent
779 with diffusive loss of radiogenic helium in nature. We do not claim to have

found the unique thermal history of the sample, but have demonstrated that plausible thermal histories reproduce our results reasonably well.

5. Conclusions

We applied the silica-glass ampoule method of Thyer et al. (2026) to conduct the first helium step heat experiment on goethite under water-saturated conditions, using proton-irradiated goethite Win-06-01b, which has previously been step heated in vacuum. The main findings are:

1. Polycrystalline goethite heated with water does not develop the contraction cracks that form in vacuum, and the water-saturated helium release differs fundamentally from release in vacuum. Published vacuum step heats of Win-06-01b follow first-order conversion to hematite, whereas the water-saturated release shows the history and time dependence of diffusion.
2. Apparent diffusivities at water saturation are approximately seven natural log units lower than those of the same sample in vacuum. Beyond 7% cumulative ^{3}He release they define a linear Arrhenius array with an activation energy of 157.1 ± 5.3 kJ/mol, and the delta plot follows multiple diffusion domain systematics. A four-domain model places 97% of the ^{3}He in a single domain.
3. Extrapolated to 25 °C, this domain is essentially quantitatively retentive of helium over billion-year timescales, supporting the use of goethite (U–Th)/He dates to date surface processes.
4. The ^{4}He/^{3}He spectrum shows that even this retentive domain lost radiogenic helium in nature, and thermally activated diffusion within the

four-domain structure can account for that loss. Steady cooling from 76–80 °C at 150–175 Ma fits the spectrum best, favoring crystallization from warm, possibly hydrothermal, fluids; burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track constraints, remains a viable alternative.

5. The water-saturated $^4He/^3He$ spectrum rises more gradually than the vacuum spectra of the same sample, suggesting that dehydroxylation and diffusion degas goethite with different geometries. Vacuum $^4He/^3He$ spectra of goethite may therefore not encode thermal histories as directly as they do for minerals that do not decompose.
6. A thermodynamic and kinetic analysis explains why water preserves goethite on experimental timescales. The water-saturated release requires solid-state conversion to be at least $10^{2.5}$ times slower than in vacuum; the roughly tenfold smaller driving force at saturation raises the barrier to topotactic hematite nucleation enough to suppress conversion far below this limit.

Fluid-mediated conversion, which transformed synthetic goethite pigment under similar conditions, would be accompanied by abrupt helium loss that is not observed. Recrystallization, however, cannot be excluded; if it contributed, the diffusivities reported here are upper bounds and the retentivity a lower bound. Separating the effects of crystallite size, chemical composition, and radiation damage on helium retention will require further water-saturated step heats of irradiated goethite of controlled grain size, which this approach makes straightforward.

Data Availability

The step heat data, permeation-lag bounds, and model outputs supporting this study are archived on Zenodo (Thyer, 2026): https://doi.org/10.5281/zenodo.22113166.

CRediT authorship contribution statement

Ben J. Thyer: Methodology, Investigation, Formal analysis, Visualization, Writing – original draft. **Jonathan Treffkorn:** Methodology, Investigation. **Kenneth A. Farley:** Conceptualization, Resources, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software programming, and with editing and LaTeX typesetting

of the manuscript. After using this tool, the authors reviewed and edited all content and take full responsibility for the content of the publication.

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